

Isaac Mubiru

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PROFESSIONAL SUMMARY

Senior Node.js engineer with 5+ years building distributed, modular backend systems — Apache Kafka topics, NATS Streaming event buses, and gRPC + Protobuf binary contracts across microservices serving four NGO tenants at github.com/zac-09. Led a zero-downtime migration of 20+ production apps from Parse/MongoDB to Firebase at MTailor, cutting ~\$5K/month in infra cost while keeping live traffic running.

Now actively studying Hypercore, Pear, and DHT architectures (Kademlia / S-Kademlia) — distributed systems is the foundation, P2P is the specialization I'm investing in next. Worldwide-remote experience: 5+ years on US-based teams from Uganda (UTC+3), heavy async written communication, CTO-direct reporting line.

CORE COMPETENCIES

Node.js	JavaScript	TypeScript	Apache Kafka	NATS Streaming	gRPC + Protobuf	Distributed Systems
Microservices	Pub/Sub	Zero-Downtime Migrations	Firebase / GCP	Docker / Kubernetes	Async-First Remote	

WORK EXPERIENCE

MTailor

Sep 2022 – Present

Software Engineer · US Remote (from Uganda, UTC+3)

- Led the end-to-end migration of **20+ production applications** from Parse/MongoDB to Firebase/GCP with **zero downtime**, reporting directly to the CTO.
- Implemented **real-time two-way sync between MongoDB and Firestore** using Node.js and Google Pub/Sub — parallel write path enabling gradual cutover.
- Built a 3D visualization feature using video overlay and ffmpeg, increasing buyer conversion.
- Migrated file storage from Amazon S3 to GCS via Python.
- Saved the company **~\$5,000/month** by migrating all services off AWS.

Dr Wealth

Aug 2021 – Jan 2022

Software Engineer · Singapore Remote

- Led optimization of a PWA delivering real-time stock market prices from a Firebase backend.
- Built ad-hoc jobs to query **over 2 million Firestore records**, keeping customer prices current from Morningstar APIs with sub-15-min freshness.
- Extended backend APIs using Firebase Cloud Functions and Express.js hosted on Heroku.

CodeBits

Jan 2020 – Jul 2021

Software Engineer · Lead of 4 · Uganda Remote

- Architected a **microservices backend with Apache Kafka, NATS Streaming, and gRPC + Protobuf** for FIDA Uganda's case management platform — independent service deploys, supported 4 NGO tenants on shared infra.
- Built a USSD legal-aid service over a flaky 2G network — Node.js backend with gRPC inter-service contracts and a retry-aware session state machine.
- Built a cross-platform mobile app for LASPNET using React Native + Expo with Firebase push notifications.
- Built a case management system for FIDA Uganda and a paralegal database for LASPNET Uganda.

Mind2matter.co

Jan 2022 – Jul 2022

Software Engineer · US Remote

- Built backends for DeFi applications using Web3 and Node.js — contract integration, transaction signing, off-chain indexing.
- Delivered React UIs from Figma designs under tight client timelines.

EDUCATION

B.Sc. Computer Engineering — [Makerere University](#)

Aug 2018 – Sep 2022

SKILLS

Proficient: Node.js, JavaScript, TypeScript, Firebase, GCP, Apache Kafka, NATS Streaming, gRPC, Protobuf, MongoDB, SQL, Python, Linux, Git, Docker, Kubernetes

Intermediate: Go, Rust, AWS, React Native, Web3

Currently studying: Hypercore, Pear, Kademlia / DHTs, libuv internals

Languages: English (native-level), Kiswahili